

LinkRAG

An Evidence Linking Layer for question answering over lectures: slides, paper and audio, connected before retrieval

This report covers everything implemented so far (Phases 1-5), how the linking layer is built — the exact formulas, algorithms, parameters and files — what was measured on the pilot lecture, the bugs found on the way, and what remains open. All numbers come from DESIGN.md, reports/ and the current code; nothing here is estimated.

Item	Value
Repository	github: m1ssmee/LinkRAG — branch main, HEAD d89cd5f
Pilot corpus	Hsieh et al., Focus (OSDI '18): 22.9-min talk audio + 27-slide deck + 19-page paper
Corpus manifest	2f3b35f27e86caf8 — 209 evidence units (111 text · 47 figure · 51 audio)
Links built	354 — 51 audio_slide · 103 figure_text · 169 deictic · 31 same_slide
Tests	250 passing (pytest, 1.4 s)
Models	bge-m3 (1024-d), faster-whisper small/int8, gpt-5.4-2026-03-05 @ temperature 0
Runtime	Python 3.11, CPU-only, macOS arm64

1. One-page summary

The problem. A lecture lives in three files: the recording, the slide deck and the paper. Existing multimodal RAG systems index each piece independently and return the top-k most similar chunks. Nothing records that a 30-second audio segment was spoken while slide 20 was on screen, or that “this graph here” refers to a specific figure. So a question whose answer is split across modalities — the speaker names a concept, the slide holds the numbers — fails, because plain similarity search never brings both halves.

The idea. Build those relations *once, at indexing time*, as typed and scored links between evidence units — the **Evidence Linking Layer**. Then (a) retrieve normally to get seeds, (b) **follow links** out from the seeds to pull in the evidence they depend on, and (c) **rerank the set** for complementarity — cover all modalities, prefer linked evidence, stop paying for repeats.

What is built. The full pipeline in both a *baseline* mode (what the papers do) and a *linkrag* mode (ours), sharing one code path so every claim is an ablation:

Phase	Component	Where	Status
1	Baseline multimodal RAG: ingest, index, hybrid retrieval, cited generation, CLIs	ingest/ index/ retrieve/baseline.py generate/	done + pilot run
2	Audio→slide alignment by monotonic dynamic programming	link/align.py	done, evaluated vs ear labels
3	figure↔text links, deictic resolution with cue tiers, same_slide, evidence graph, optional VLM captions	link/figure_text.py deictic.py graph.py ingest/vlm_caption.py	done, evaluated
3b	Paper added; caption-anchored + vector-cluster figure extraction	ingest/pdf.py	done
4	Link-following retrieval; P1-style iterative baseline; composed linkrag_iter	retrieve/linkrag.py iterative.py	done, 4-mode comparison
5	Complementarity-aware set reranking (+ MMR ablation, optional cross-encoder)	retrieve/rerank.py	done, mode×rerank matrix
—	Corpus manifest hashing, regression harness, measurement rules	manifest.py scripts/	done

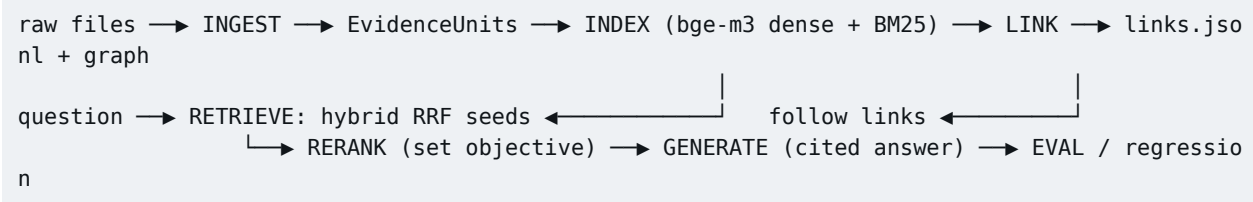
Headline measurements (pilot01, n small — indicative, not results).

- Slide alignment: **76.2 %** exact-slide accuracy vs 57.1 % for the P2-style per-segment argmax; 0 backward steps vs 9; holds a single slide through 4 minutes of Q&A where the baseline wanders across 5.
- Deictic resolution (“this”, “here” → which figure): precision **79 %** on 24 pairs, detection recall **86 %**, 0 false positives on the known-negative window; tier-2 cues 86 % vs tier-3 70 %.
- Retrieval, 4 questions, k=8: baseline 33.3 % recall → link-following 41.7 % (0 LLM calls) → P1 iterative 47.9 % (4 LLM calls) → **both composed + complementarity rerank 70.8 % ± 7.2**, with every evidence set spanning all three modalities.

The honest headline for a mentor: alignment and deixis are clearly ours and measured against ear labels. On end-to-end retrieval, P1's re-querying alone beats link-following alone on this 4-question set; the two fix different failures (Q4 needs an *edge*, Q3 needs a *seed*), and their composition is what wins. n=4 and a stale gold set mean nothing retrieval-side is paper-ready yet.

2. System overview

Every stage consumes and produces the same currency: an EvidenceUnit (id, modality, text, embedding, Location = file + page/bbox or file + start/end seconds) and a Link (src_id, dst_id, link_type, score, metadata). Unit ids double as citations (hsieh:a28, osdil8_slides_hsieh:p20:t0).



Stage	baseline mode (papers)	linkrag mode (ours)
ingest	text chunks, figures, audio windows extracted independently	identical extraction; bboxes, captions, word timestamps recorded for the linker
index	bge-m3 embeddings, exact dense search (numpy), BM25	same index
link	n/a — no links exist	audio_slide, figure_text, deictic, same_slide → networkx MultiDiGraph
retrieve	RRF fusion of dense + BM25, plain top-k	top-k as seeds → 1-hop link expansion → set rerank
generate	cited answer, OpenAI-compatible endpoint	same, plus modality tags in the prompt

Rule enforced in code: `ask.py --mode linkrag` errors out if no links exist rather than silently falling back to baseline, because a silent fallback fakes the ablation.

3. Ingestion — how evidence units are made

3.1 Documents (PyMuPDF)

- **Text:** page blocks are packed into ~300-token chunks with 60-token overlap; each chunk keeps the union bbox of its blocks so it can be placed on the page later.
- **Raster figures:** `page.get_images()`, filtered by `min_figure_area_px = 10 000`, OCR'd with `pytesseract` (slide decks caption nothing, so the image text is the only content).
- **Deck detection:** a page aspect ratio above 0.6 (landscape) marks a slide deck; the paper is portrait. Per-file override via `ingest.slide_deck_files`. Deck-ness drives which figure extractor runs and whether `same_slide` links are created.
- **Caption-anchored figures (papers):** LaTeX papers draw plots as vector operators, invisible to `get_images()` — the Focus paper has 13 captioned figures and yielded 2 logo fragments. Fix: find blocks matching `^(Figure|Table) N[:.]`, take the region above the caption bounded by the nearest *prose* block in the same column (a block of ≥ 12 words; `BODY_TEXT_WORDS`), height clamped to 40–420 pt, render it. Bounding at the nearest *any* block collapsed 9 of 14 regions to 6–16 pt because a plot's own axis labels are text blocks — that was the first attempt and it was wrong.
- **Vector-cluster figures (decks):** merge every `get_drawings()` and `get_images()` bbox whose gap is < 12 pt into clusters, to a fixpoint. Drop the page backdrop (≥ 80 % of the page), clusters under 8 000 pt², corner logos (≤ 15 000 pt² within 100 pt of a corner), and *template*

chrome — an identical bbox appearing on ≥ 3 pages (21 title banners dropped on pilot01). Result: deck figures went 18 $\rightarrow$ 31 and slides 13/16/20/22/23 gained their plots. These are new extraction parameters, not retuned thresholds.

3.2 Audio (faster-whisper)

- **Transcription:** small model, int8, CPU, word-level timestamps. 22.9 min of audio took 476 s ($\approx 2.9\times$ realtime) on the plain run.
- **Slide-guided ASR:** ~ 75 terms mined from the deck go into whisper's `initial_prompt`. Effect: the talk's central term *ingest* went from 7 correct / 20 “interest” to 16 / 10; but 4 spoken “top-K” became “type -k”. A hypothesis that the hyphen caused it was tested by a full re-run — identical output — and rejected. The regression is open.
- **Segmentation:** whisper segments are decoder chunks, not sentences (only $\sim 26\%$ end on punctuation). Packing whole segments cut 74 % of boundaries mid-sentence — *worse* than fixed 30-s windows (70 %). Splitting the *word stream* on terminal punctuation and packing sentences to a 30-s budget gives **0 %** mid-sentence cuts. `audio_segmentation: sentence | fixed` keeps the papers' behaviour as an ablation.
- **Frozen transcript:** `data/processed/transcripts/hsieh.frozen.json` (guided v2) is used verbatim by every later phase; whisper never re-runs on pilot01. 51 audio units.

3.3 Index

bge-m3 embeddings (1024-d, L2-normalised) stored as a numpy matrix; dense search is an exact inner product (8.8 ms/query at $100\text{ k} \times 1024$ measured). BM25 (`rank_bm25`) is rebuilt on load. faiss was removed: faiss-cpu and torch each bundle an OpenMP runtime and the pair segfaults on macOS arm64; the documented workaround is documented upstream as possibly giving silently wrong results. Hybrid retrieval fuses the two rankers by Reciprocal Rank Fusion, $score(u) = \sum 1 / (60 + rank_i(u))$, 50 candidates from each ranker.

4. The Evidence Linking Layer — exact technicalities

Four link types are built at index time by `scripts/build_links.py` and written to `data/processed/links.jsonl` (first line is a `_meta` record carrying the corpus manifest hash; loading links against a different corpus raises `LinkManifestMismatch` instead of silently degrading to top-k). Each module has a baseline mode that is the ablation.

link type	from → to	signal	count	mean score
audio_slide	audio segment → deck page text	monotonic DP over a fused similarity matrix	51	0.674
figure_text	figure → text chunk (any page, any document)	numbered reference, layout gap, page prior, cosine, IDF overlap	103	0.652
deictic	audio segment → figure on the aligned slide	cue phrase × tier, slide match, cosine on cue context, IDF overlap	169 (raw cue hits)	0.632
same_slide	figure → its own slide's text	co-location on a deck page (certain)	31	1.000
			354	

4.1 audio_slide — monotonic audio-to-slide alignment (link/align.py)

Input. Audio units $a_1 \dots a_n$ in time order ($n = 51$) and slide text units $s_1 \dots s_m$ in page order ($m = 27$) from a *separate* PDF. Decide which slide each segment discusses.

Step 1 — similarity matrix. For every (segment, slide) pair:

$$S_{ij} = w_d \cdot \cos(e_i, e_j) + w_b \cdot \widehat{BM25}(a_i, s_j) + w_k \cdot K_{ij}, \quad K_{ij} = \frac{\sum_{t \in a_i \cap s_j} \text{idf}(t)}{\sum_{t \in s_j} \text{idf}(t)}$$

e are bge-m3 embeddings; BM25 is min-max rescaled per row to $[0, 1]$ because raw BM25 is unbounded and cannot be summed with a cosine; K_{ij} is an IDF-weighted rare-term overlap so sharing “NoScope” counts far more than sharing “the”. Weights: $w_d = 0.6$, $w_b = 0.25$, $w_k = 0.15$.

Step 2 — the structural prior. A lecturer walks *forward* through a deck, so the assignment should be (near-)monotonic in time and each segment's slide is constrained by its neighbours, not only by its own score. Let D_{ij} be the best total score of aligning segments $1..i$ with segment i on slide j . With forward-jump penalty λ , per-skipped-slide penalty σ , back-jump penalty β and back-jump limit B :

$$D_{1j} = S_{1j} - \sigma j - \mu j$$

$$D_{ij} = S_{ij} + \max \left\{ D_{i-1,j}, D_{i-1,j-1}, \max_{j' \leq j-2} [D_{i-1,j'} - \lambda(j-j') - \sigma(j-j'-1)], \max_{1 \leq b \leq B} [D_{i-1,j+b} - \beta b] \right\}$$

$$\text{answer} = \max_j [D_{nj} - \sigma(m-1-j)]$$

The four cases are: stay on the slide; advance one; jump forward skipping slides; jump back up to B slides. The σ terms at the head and tail charge for slides never assigned before the first and after the last segment, so the penalty counts *every* unassigned slide. μ is a start prior pulling segment 1 toward slide 1 (without it the certain title-slide segment was assigned p3). Values: $\lambda =$

0.05, $\sigma = 0.02$, $\beta = 0.15$, $B = 2$, $\mu = 0.02$. $B = 0$ gives strict monotonicity; $B = 2$ with $\beta > \lambda$ lets a speaker flip back to answer a question without forcing the path to absorb that error elsewhere. Backtracking pointers recover the path.

Step 3 — why it is $O(nm)$ and not $O(nm^2)$. The forward-jump case is a max over all $j' \leq j - 2$, which looks quadratic. But the penalty is affine in $(j - j')$, so

$$D_{i-1,j'} - (\lambda + \sigma)(j - j') + \sigma = [D_{i-1,j'} + (\lambda + \sigma)j'] - (\lambda + \sigma)j + \sigma$$

and the bracketed term admits a running prefix maximum computed in one left-to-right sweep. The back-jump case is a max over $B = 2$ terms. Both are $O(1)$ amortised per cell.

`tests/test_align.py` checks the shipped DP against a brute-force $O(nm^2)$ reference on random matrices — an off-by-one in the prefix max would return subtly wrong paths in silence.

Baseline. `align_naive`: $\text{argmax}_j S_{ij}$ per segment, no sequence structure — roughly what P2 does. Selected by `link.align.method`: `naive`.

Output. One `audio_slide Link` per segment with score S_{ij} on the chosen path; `links.npz` caches S and the path so plotting and evaluation skip re-embedding.

Measured against ear labels v2 (42 talk segments + 9 Q&A segments, ± 3 s)

	monotonic DP	naive argmax (P2-style)
exact slide (n = 42)	32/42 = 76.2 %	24/42 = 57.1 %
within ± 1 slide	92.9 %	73.8 %
exact, non-ambiguous subset (n = 33)	72.7 %	57.6 %
backward steps	0	9
distinct slides covered (of 26 showable)	21	15
Q&A (n = 9): distinct slides assigned	1 — holds slide 26	5 — wanders

11 of the DP's 13 errors are one slide *ahead* of truth (p5→p6, p8→p9, p13→p14, p17→p18) — the speaker talks about the next slide before advancing. A lag term is the obvious next parameter; not attempted. μ was fitted on the same labels it is scored on, so 76.2 % is optimistic. Correcting the labels (v1 → v2) alone moved accuracy 69.0 % → 76.2 % with no code change — a reminder to separate label quality from method quality before claiming credit.

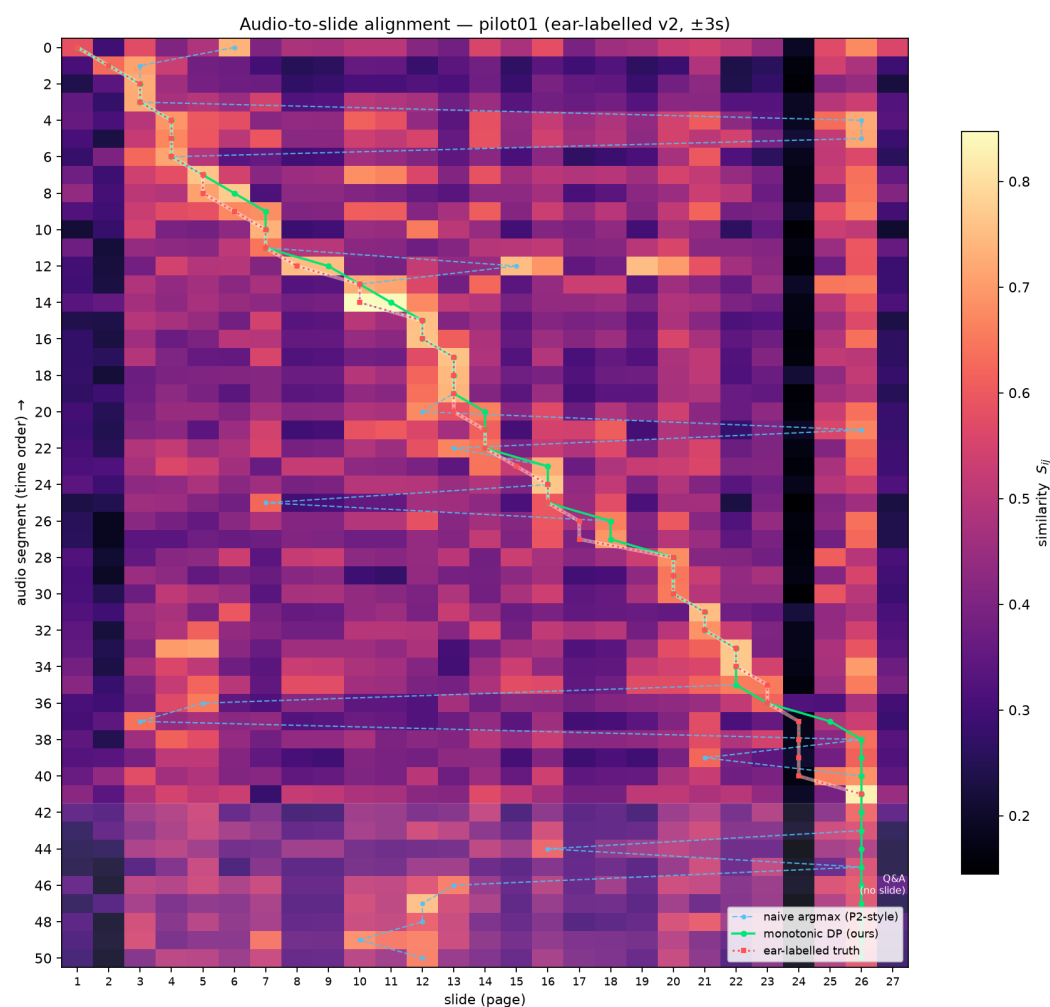


Figure 1 — similarity matrix S with the DP path, the naive argmax and the ear-labelled truth overlaid. The naive path's errors are jumps across the deck; the DP's are adjacent slides.

4.2 figure_text — figure to explaining text (link/figure_text.py)

P3 treats pages as independent, so a figure and the paragraph explaining it three pages later never meet. Here candidates are scored across the *whole corpus*, with page distance as a soft prior. For figure f and text unit t :

$$F_{ft} = w_r R_{ft} + w_l L_{ft} + w_p P_{ft} + w_d \cos(e_f, e_t) + w_o O_{ft}$$

term	what it is	how computed	weight
R (reference)	text explicitly names the figure	Figure N/Fig. N/Table N regex, <i>kind-aware</i> (“Table 2” cannot match “Figure 2”), restricted to the same or an adjacent page (reference_page_window = 1); plus unnumbered descriptive refs (“the diagram below”) resolved by bbox direction	0.30
L (layout)	vertical whitespace between the two bboxes on the same page	gap in points, linearly decayed to 0 at 220 pt; 0 if on different pages	0.15
P (page prior)	distance in pages	$\exp(- \text{page}(f) - \text{page}(t) / \tau)$, $\tau = 2$ — a prior, never a filter	0.15
cos	semantic similarity	bge-m3 on caption + OCR (+ optional VLM caption) vs chunk text	0.25
O (overlap)	rare shared terms	IDF-weighted overlap, same form as K_{ij} above	0.15

Threshold 0.50; at most 8 links per unit. Every link stores *why* it was made in metadata (which signal fired), because a link the retriever cannot explain is a link the evaluation cannot score.

Baseline: same-page candidates only — the best a page-independent pipeline can do; same code path, so the ablation isolates cross-page reach alone.

Measured

signal (paper, 19 pp.)	links	mean score
explicit numbered reference	31	0.836
explicit descriptive (“the diagram below”)	0	—
layout proximity	21	0.539
semantic only	0	—
paper total	52	

Corpus-wide there are now 103 figure_text links; 7 cross a page boundary and 10 cross documents (9 deck-figure → paper-text, 1 paper-figure → deck-text) — none of which the baseline can produce. The deck alone gave 5 links and 0 cross-page, because slides have no captions and never write “Figure N”: this mechanism was designed for papers and the paper is where it fires. A cross-document figure↔figure pass was *rejected*: the top pairs at 0.92 were all empty-content figures embedding a provenance placeholder; on content-bearing figures the max cosine was 0.546 and no link was emitted.

4.3 deictic — “as you can see here” → the figure (link/deictic.py)

The referent of *here* is on a slide in a different file and nothing in the words identifies it. This module resolves it by **composing with 4.1**: the segment is already aligned to a slide, so the candidate set shrinks from “every figure in the corpus” to “the figures on the slide that was on

screen”. Deixis is resolved by sequence structure, not by the deictic word.

Cue detection. Cue phrases are generated as determiner × object-noun × direction ({the, this, that} × {arrow, box, step, line, circle, node, column, row, bar, point, block, region, curve, axis} × {above, below, on the left, on the right, here, there}) plus a 14-phrase base list — 266 phrases from two short config lists. They are compiled into one regex alternation, longest phrase first, so “as you can see” is not also reported as “see”/“this”. Word timestamps locate each cue and its context window.

Tiering. Each cue gets a tier that multiplies the cue weight, so tier changes ranking without moving any threshold:

tier	meaning	weight	pilot01 distinct pairs
1	explicit object deixis (“this arrow here”)	1.0	0 — this speaker never does it
2	bare pronoun + a visual word in the same sentence (“you can see this”)	0.6	14 scored
3	bare pronoun alone (“this”, “that”)	0.3	10 scored

Scoring. For segment a carrying cue c and figure v :

$$D_{av} = w_s [page(v) = slide(a)] + w_c tier(c) + w_d \cos(e_{ctx(a,c)}, e_v) + w_o O_{av}$$

$ctx(a, c)$ is the words around the cue rather than the whole 30-s segment; O_{av} is IDF overlap against the figure's OCR/caption — the tie-break when one slide carries several figures. Weights: $w_s = 0.45$, $w_c = 0.15$, $w_d = 0.20$, $w_o = 0.20$; scores are normalised by the earnable weight mass.

Known property, documented rather than hidden: in linkrag mode every scored candidate is on the aligned slide, so every candidate starts at $w_s/mass = 0.45$ — exactly the threshold. Measured floor over the pilot links is 0.56. The emitted set is therefore effectively “every cue on a slide that has a figure”, capped at 8 per unit. It was deliberately not retuned, because retuning would change every recorded deictic number mid-project. Raw links double-count (one segment can fire six cues at one figure), so reporting leads with **distinct (segment, figure) pairs at max score**.

Baseline: no alignment — every figure in the corpus is a candidate, ranked by similarity alone. A first version of this ablation was invalid: it still added $w_s \cdot on_slide$, i.e. used the alignment it was meant to be measured without, and returned the identical 18 links. Guarded now by `test_baseline_does_not_use_the_alignment`.

Measured against 10 ear-labelled pointing windows

	value	n / note
windows scorable	8 / 10	was 3/10 before vector-cluster figure extraction; remaining 2 = a table (no table extraction) and the live demo
precision on positives	79 %	24 pairs — tier 2: 86 % (14), tier 3: 70 % (10)
detection recall over scorable windows	86 %	12 / 14 segments inside a window produced a pair
false positives, slide-24 external demo	0	the demo is not in the deck, so any link there would be wrong

Tier 2 beating tier 3 on referent-level ground truth is the first evidence that tiering does work. The binding constraint moved from “no figure units to point at” to the two structurally unresolvable windows.

4.4 same_slide, the evidence graph and corpus binding

- **same_slide** (link/figure_text.py): each figure on a deck page links to that page's text at a fixed score 1.0. Co-location is certain, not scored, so it does not compete inside figure_text where a slide could never earn caption or reference evidence. Ablatable (link.same_slide.enabled). Correctly produced 0 links from the portrait paper.
- **Graph** (link/graph.py): a `networkx.MultiDiGraph` over all units and links. `neighbors(unit_id, link_types, min_score, direction)` dedupes parallel edges to the best score and walks both directions; `subgraph_around` and `GraphML` export feed `scripts/plot_graph.py`. Edges were once keyed by `link_type`, which silently collapsed 19 of 41 deictic links; caught only because the per-type counts did not sum to the total. Edges are auto-keyed now.
- **Manifest** (manifest.py): a hash over the corpus files (timestamp excluded, so rebuilds do not invalidate it) stamps `links.jsonl` and the regression gold set. A mismatch is a hard error for links and a printed warning for gold — the current gold is stamped `dded007ab45f5b9f` (2-document corpus) against the live `2f3b35f27e86caf8`, and every run says so.

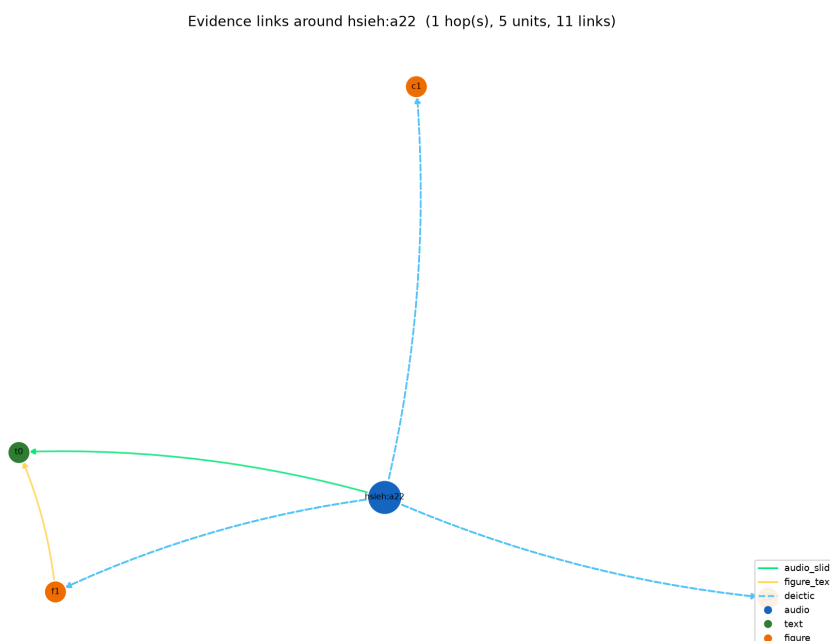


Figure 2 — the 1-hop neighbourhood of audio segment `hsieh:a22` in the evidence graph: its aligned slide, the figures it points at, and the paper text those figures link to.

5. Link-following retrieval (retrieve/linkrag.py)

Top-k becomes **seeds** ($k_{\text{seed}} = 5$ from hybrid RRF). The graph is then traversed 1 hop along {audio_slide, figure_text, deictic} and the best $k_{\text{final}} = 8$ are kept. A unit enters the set because something already retrieved *depends on it*, not because it independently resembled the query — which is how a slide sharing no vocabulary with the question can be retrieved. An expanded unit scores

$$S_{\text{expanded}} = S_{\text{seed}} \cdot W_{\text{link}} \cdot \gamma, \quad \gamma = 0.5$$

Scale problem and fix. RRF seed scores are ~0.03 while link weights are 0.6–1.0, so seed $\times$ link $\times \gamma$ (~0.01) sits two orders below any seed and expansion can only fill the tail.

normalise_seeds rank-normalises seeds to (0, 1] (rank r of $n \rightarrow (n - r + 1)/n$) so an expanded unit can outrank a weak seed. Off by default because it changes every recorded number.

Provenance. Every RetrievedUnit records seed-or-expanded, the seed and the link type that brought it in (explain()); expansion_report distinguishes “seeds had edges but nothing passed the filter” from “graph empty or stale”, since both make linkrag look identical to baseline.

Baselines. baseline = plain top-8. iterative (retrieve/iterative.py) is our P1 approximation: retrieve, ask the LLM for one follow-up query, retrieve again, merge (2 rounds, 8 per round — 1 LLM call per question on the critical path). linkrag_iter seeds link-following from the iterative retriever, composing the two.

Four modes, 4 questions, $k = 8$, normalise_seeds on

mode	recall@8	precision@8	LLM calls	Q1	Q2	Q3	Q4
baseline (top-k)	33.3 %	9.4 %	0	0	100	0	33
iterative (P1)	56.2 %	18.8 %	4	0	100	25	100
linkrag	50.0 %	15.6 %	0	0	100	0	100
linkrag_iter	75.0 %	28.1 %	4	0	100	100	100

Why they compose. Q4 needs an *edge*: hsieh:a28 $\rightarrow$ audio_slide (0.681) $\rightarrow$ slide p20, whose text carries “Optimize for Ingest Cost / Balance / Optimize for Query Latency”. That slide had been the missing gold unit in every regression since Phase 1 and no re-querying found it; link-following brought it in and the generated answer then named the three configurations citing p20. Q3 needs a *seed*: the edge hsieh:a0 $\rightarrow$ p1:t0 exists, but hsieh:a0 never enters the top-8 (all Q3 seeds are paper text), so traversal never starts there; a second query supplies the entry point. **Link-following inherits the seed retriever's recall and cannot reach a region of the graph that has no seed.**

Q1 is 0 % in every mode partly because its gold locator is a slide the retriever never reaches; Q3's original design intent (“split across audio and slide”) no longer holds since the paper made it answerable from one page — kept and flagged, not rewritten.

6. Complementarity-aware reranking (retrieve/rerank.py)

Top-k scores each unit in isolation, so eight near-identical audio segments score as well as a set that also carries the slide they describe. This module scores the **set**. Given candidates C (pool =

20) with relevance $rel(u)$, choose $R \subseteq C$, $|R| = 8$, maximising

$$F(R) = \sum_{u \in R} rel(u) + \alpha |\{m(u) : u \in R\}| + \beta |\{(u, v) \in R^2 : u \rightarrow v \in E\}| - \gamma \sum_{u \neq v, m(u) = m(v)} \cos(e_u, e_v)$$

$m(u)$ is the modality, E the linking layer's edges. Read: cover more modalities ($\alpha = 0.3$ per new modality), prefer evidence linked to what is already selected ($\beta = 0.2$ per edge), stop paying for a unit that restates one already chosen ($\gamma = 0.4$). Redundancy is charged **only within a modality** — an audio segment and the slide it describes are *supposed* to be similar. Exact maximisation is NP-hard (facility-location-style), so selection is greedy from $\arg\max rel$, adding the largest marginal gain:

$$\Delta(u | R) = rel(u) + \alpha [m(u) \notin m(R)] + \beta |\{v \in R : u \leftrightarrow v \in E\}| - \gamma \sum_{v \in R, m(v) = m(u)} \cos(e_u, e_v)$$

Ablations on one code path: none (plain top-k), mmr (standard MMR, $\lambda \cdot rel - (1-\lambda) \cdot \max \cos$, $\lambda = 0.5$ — modality-blind and link-blind, isolating what α and β add), complementarity. An optional cross-encoder (e.g. BAAI/bge-reranker-base) can supply $rel(u)$ from a modality-tagged string ([AUDIO 18:40-19:10] ..., [FIGURE p.12] ...); a missing model degrades to retrieval scores with a warning. Tests include a 5-near-duplicate-audio + 1-linked-figure candidate list where complementarity must admit the figure and top-k must not.

mode × rerank matrix — smoke test, n = 4, 3 repeats for LLM-touching cells

mode	rerank	recall@8	precision@8	modalities	redundancy
baseline	none	33.3 %	9.4 %	1.75	0.711
baseline	complementarity	12.5 %	3.1 %	2.25	0.682
iterative	none	47.9 ± 7.2 %	16.7 ± 3.6 %	2.50	0.709
iterative	complementarity	29.9 ± 2.4 %	11.5 ± 1.8 %	2.50	0.649
linkrag	none	41.7 %	12.5 %	2.75	0.688
linkrag	complementarity	37.5 %	12.5 %	2.75	0.635
linkrag_iter	none	63.2 ± 4.8 %	22.9 ± 1.8 %	3.00	0.716
linkrag_iter	mmr	54.2 ± 0.0 %	21.9 ± 0.0 %	3.00	0.568
linkrag_iter	complementarity	70.8 ± 7.2 %	27.1 ± 1.8 %	3.00	0.599

The interaction finding. Complementarity raises modality coverage and lowers redundancy in every mode — the objective does what it was written to do — but it improves recall only in the one cell whose pool already spans all three modalities (linkrag_iter, +7.6 pp) and hurts everywhere else (baseline −20.8 pp). On a mostly-text pool, “add an unrepresented modality” means evicting a gold text unit to admit a figure. α and β are worth paying only when the pool contains cross-modal evidence to select. The monotone trend across pool coverage (1.75 → 3.00) is the observation to re-test on a larger question set.

7. Generation, evaluation harness and measurement discipline

7.1 Generation (generate/answer.py)

- Any OpenAI-compatible /v1/chat/completions endpoint (Ollama or OpenAI). Model pinned to the dated snapshot gpt-5.4-2026-03-05; gpt-5.5 was rejected because it only accepts temperature 1.
- temperature: 0.0 and seed: 20260910 are sent in the request body (asserted by a captured-request test). The max_tokens / max_completion_tokens drift is auto-detected from the 400 response. requests.RequestException (not just ConnectionError) is caught after a ReadTimeout killed a batch run.
- Evidence is presented with modality tags and unit ids; answers must cite ids. An id-level citation check exists, and Phase 1 already showed why it is insufficient: Q3 cited a real unit and was still wrong (an audience member's self-introduction shared a unit with the closing line).

7.2 Harness

- tests/regression/pilot01_questions.jsonl — Q1 slides-only, Q2 audio-only, Q3 cross_modal_split, Q4 cross_modal_deictic, with gold matched by **file + page / file + time overlap, never by unit id** (audio ids regenerate whenever chunking changes). run_regression.py appends every run to reports/regression.md.
- compare_retrieval.py runs the mode x rerank matrix, prints per-question recall/precision, modality distribution and redundancy, runs LLM-touching cells 3x and **refuses to print a single-run number** for them.
- eval_alignment.py / eval_deictic.py score against the ear labels in data/labels/pilot01/ (slide timeline v2, pointing windows).
- Standing instrument reported every phase: the modality distribution of each retrieved set — a gain that only reshuffles within one modality is not the cross-modal gain this project claims.

7.3 Measurement rules (binding, in DESIGN.md)

- Latency counts only if measured on an idle machine, single process, warm model, ≥ 3 repeats, reported as the median. Exists because an early draft claimed guided ASR “costs 2.1x decode time” from a run that was sharing cores with an embedding job.
- An ablation that scores *exactly* equal to the full system, or *exactly* zero, is presumed a bug until inspected. Both happened (deictic baseline, twice).
- A table whose per-category counts do not sum to the total is reconciled before reporting. This caught the graph edge collapse.
- Every new script or code path is executed once for real, not mocked, before it counts as done. Three defects shipped with a green test suite before this rule: a NameError on a never-run manifest write, --mode linkrag silently running baseline retrieval, and the faiss/torch crash a stub encoder never triggered.
- LLM determinism is not assumed: with temperature 0 and a seed, both LLM-touching modes gave 3 different results in 3 runs (system_fingerprint is null for this account). Hence mean $\pm$ std, and one cell has a 12 pp std — larger than most differences anyone would want to claim.

7.4 Bugs found, fixed and guarded

bug	symptom	fix / guard
BM25 tie-break	argsort[:, -1] broke ties in reverse corpus order; last-ingested unit ranked first on no-match queries	stable descending sort; same class fixed in figure_text and deictic; regression test
Graph edge collapse	edges keyed by link_type; 19/41 deictic links dropped	auto-keyed edges; neighbours dedupe to best score
Deictic ablation leak	baseline used the alignment it was measured against; then scored exactly 0 by construction	slide term removed; scores normalised by earnable mass; test
Mode flag not wired	run_regression --mode linkrag only changed the prompt suffix	retrieve_for_mode dispatch; mocked-retriever tests
Vector figures invisible	paper: 2 figures of 13; deck: plots on slides 13/16/20/22/23 missing	caption-anchored regions (papers); drawing/raster clustering with template removal (decks)
Segment packing	sentence-aware packing cut more mid-sentence (74 %) than fixed windows (70 %)	split the word stream on terminal punctuation → 0 %
Timeout uncaught	ReadTimeout killed a batch	catch RequestException
faiss + torch	OpenMP double-load segfault	faiss removed; exact numpy search

8. How this differs from the three baseline papers

capability	P1 · MI-RAG (ICLR 26)	P2 · Intra-Video Temporal RAG (CMC 26)	P3 · MARA (ACM MM 25)	LinkRAG
modalities	text	audio + video frames	text + figures (documents)	text + figures + audio
relation between units	none	timestamp co-occurrence only	none (pages independent)	4 typed, scored link types
crosses file boundaries	no	no — stops at the video	no	yes: audio→deck, deck→paper
how the evidence set grows	LLM re-queries (more fuzzy retrieval)	—	—	traverses saved links; 0 LLM calls
set-level selection	—	—	—	modality / link / redundancy objective
faithfulness check	—	none	—	id-level citation check; claim-level needed

P1 — MI-RAG

Limit: re-querying raises recall but each round is another fuzzy retrieval; nothing models *how* two chunks relate. **We do:** expansion along a known typed relation, which can admit a slide sharing no vocabulary with the question. **Measured:** we implemented P1 faithfully; alone it beats link-following on this set (56.2 vs 50.0 % recall) at 1 LLM call per question vs 0; composed with link-following and set reranking it reaches 70.8 %. Each fixes a failure the other cannot.

P2 — Intra-Video Temporal-Aware RAG

Limit: audio and visuals are fused by clock time *inside one video*; a recording plus a separate slide PDF is exactly the case it cannot handle, and it has no answer-faithfulness check. **We do:** align 51 audio segments to a 27-page deck in another file using the forward-walk structure of a lecture. **Measured:** 76.2 % vs 57.1 % exact-slide accuracy against the P2-style per-segment argmax, 0 vs 9 backward steps, 1 vs 5 slides during Q&A.

P3 — MARA

Limit: pages are independent, so a figure on page 3 and its explanation on page 6 never connect; no audio. **We do:** whole-corpus figure↔text scoring with explicit numbered references, layout, page prior and semantics — plus a deictic link type that a text-only system cannot have. **Measured:** 103 figure_text links, 7 cross-page and 10 cross-document, none reachable by the same-page baseline.

9. Open items and what is not claimed

- **n = 4 questions.** 24 further questions are proposed in reports/question_proposal_pilot01.md and await ear-verification before entering the regression set. No retrieval number is paper-ready.
- **Gold set is stale** (stamped for the 2-document corpus). A re-stamped proposal is in reports/gold_proposal_pilot01.md, unapplied pending approval.

- **Reranker** needs ablation of α , β , γ on recall over a reliably cross-modal pool and a larger question set before keeping or dropping it.
- **Alignment**: μ fitted on the scored labels (no held-out split); a lag term for the speaker-runs-ahead pattern is untried.
- **Deictic threshold** is vacuous in linkrag mode (documented); a referent-labelled dev set is needed to trade recall for precision honestly.
- **ASR**: the “top-K $\rightarrow$ type -k” regression from the vocabulary prompt is unexplained; a query-time *ingest* $\leftrightarrow$ *interest* alias was considered and rejected as lecture-specific.
- **Discourse boundaries**: units can span a speaker change (the Q3 false-author case); no turn detection attempted.
- **Faithfulness**: a claim-level check is required; the id-level check is known to pass a wrong answer.
- **UI** (src/linkrag/ui/) and the full eval stage are unbuilt.

Appendix — repository map

```
src/linkrag/
  core.py          EvidenceUnit, Link, Location, Mode, load_config, stage_timer
  manifest.py      corpus hashing; gold/link stamps
  ingest/ pdf.py docx.py image.py audio.py vlm_caption.py
  index/  numpy dense + BM25, persisted to data/processed/index/
  link/   align.py figure_text.py deictic.py graph.py
  retrieve/ baseline.py linkrag.py iterative.py rerank.py
  generate/ answer.py (OpenAI-compatible, cited answers)
scripts/ ingest.py build_links.py ask.py run_regression.py compare_retrieval.py
          eval_alignment.py eval_deictic.py make_alignment_labels.py plot_alignment.py plot_graph.py
tests/    250 tests incl. brute-force DP reference, ablation-leak guards, captured-request assertions
reports/  regression.md phase1_pilot.md phase2_step0_asr.md deictic_eval_pilot01.md audit_phase4.md ...
data/labels/pilot01/ slide timeline v2, pointing windows (ear-labelled)
configs/default.yaml every weight, penalty and threshold quoted in this report
```